THE HENSELIZATION AND STRICT HENSELIZATION OF A P-RING ARE P-RINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The henselian pair of a P-ring is a P-ring, and it suffices to check property P on formal fibres. Lemma \ref{lemma-henselization-pair-P-ring} and Lemma \ref{lemma-check-P-ring-maximal-ideals}, cited at proof lines 13162 and 13165-13166. Give the henselization case and reduce the strict henselization case to formal fibres.
- The fibre of the strict henselization at a prime is a product of residue fields of the primes over the prime below. Lemma \ref{lemma-fibres-henselization}, cited at proof lines 13171-13173. Turns the product of formal fibres into the single tensor product where the completion of the strict henselization appears.
- The completion of the henselization is formally smooth over the completion of the ring, hence regular. Lemma \ref{lemma-henselization-noetherian} and Proposition \ref{proposition-fs-regular}, cited at proof lines 13180-13183. Makes the base change of property P along the completion available through condition (C).
- Separability of the residue field extension of the strict henselization. condition (E) and Lemma \ref{lemma-fibres-henselization}, used at proof lines 13193-13195. Transfers property P from the residue field of the prime below to the residue field of the prime.

Source reference. Original lemma 0C36: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15181–15181. The lemma's own canonical tag,

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quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0BKA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14553–14553. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BIU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14502–14502. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07QQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9840–9840. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 06LJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8468–8468. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9802–9802. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian local ring which is a P-ring where P satisfies (B), (C), (D), and (E). Then the henselization R^h and the strict henselization R^{sh} are P-rings.*

Proof. We have seen this for the henselization in Lemma 0BKA. To prove it for the strict henselization, it suffices to show that the formal fibres of R^{sh} have P, see Lemma 0BIU. Let $\mathfrak{r} \subset R^{sh}$ be a prime and set $\mathfrak{p} = R \cap \mathfrak{r}$. Set $\mathfrak{r}_1 = \mathfrak{r}$ and let $\mathfrak{r}_2, \ldots, \mathfrak{r}_s$ be the other primes of R^{sh} lying over $\mathfrak{p}$, so that $R^{sh} \otimes_R \kappa(\mathfrak{p}) = \prod_{i=1,\ldots,s} \kappa(\mathfrak{r}_i)$, see Lemma 07QQ. Then we see that

$$\prod\nolimits_{i=1,\ldots,t} (R^{sh})^\wedge \otimes_{R^{sh}} \kappa(\mathfrak{r}_i) = (R^{sh})^\wedge \otimes_{R^{sh}} (R^{sh} \otimes_R \kappa(\mathfrak{p})) = (R^{sh})^\wedge \otimes_R \kappa(\mathfrak{p})$$

Note that $R^\wedge \to (R^{sh})^\wedge$ is formally smooth in the $\mathfrak{m}_{(R^{sh})^\wedge}$-adic topology, see Lemma 06LJ. Hence $R^\wedge \to (R^{sh})^\wedge$ is regular by Proposition 07PM. We conclude that property P holds for $\kappa(\mathfrak{p}) \to (R^{sh})^\wedge \otimes_R \kappa(\mathfrak{p})$ by (C) and our assumption on R. Using property (B), using the decomposition above, and looking at local rings we conclude that property P holds for $\kappa(\mathfrak{p}) \to (R^{sh})^\wedge \otimes_{R^{sh}} \kappa(\mathfrak{r})$. Since $\kappa(\mathfrak{r})/\kappa(\mathfrak{p})$ is separable algebraic, it follows from (E) that P holds for $\kappa(\mathfrak{r}) \to (R^{sh})^\wedge \otimes_{R^{sh}} \kappa(\mathfrak{r})$. $\square$